Manifold-Aware Concept Erasure and Evaluation

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Abstract

Although the ambient representation space of a language model is high-dimensional, the set of sentence representations used in practice can lie on a much lower-dimensional natural manifold. This creates a problem for gradient-based concept erasure: when erasure updates are computed in the full ambient space, they are likely to push representations off that manifold and damage control concepts instead of removing only the target one. We introduce Tangent Gradient-based Concept Erasure (TanGCE), which constrains nonlinear erasure updates to a locally estimated tangent space so that the intervention stays aligned with the geometry actually used by the model. This geometric constraint lets TanGCE safely employ larger erasure strengths λ than ambient updates, improving removal of nonlinear concept information while keeping control-task changes small. Across three text datasets—sycophancy, gender, and safety—in Qwen2.5-0.5B, and a large vision benchmark with 40 facial attributes in CLIP ViT-B/32 representations of CelebA, TanGCE achieves the strongest erasure precision together with competitive control-concept preservation. These results support a simple picture: once concept removal is constrained to the natural manifold of representations, stronger interventions become possible without paying the same collateral-damage cost.

1 Introduction

Effective concept erasure in neural representations requires a delicate balance between removing targeted information and preserving the rest of the representation. This is especially difficult for nonlinear concepts: an edit may defeat one trained probe while leaving behind signal that a freshly trained nonlinear classifier can still recover, as emphasized by Elazar and Goldberg's cautionary analysis of post-hoc recovery after adversarial removal [Elazar and Goldberg, 2018]. For erasure to be useful, it must therefore achieve two goals simultaneously: high *erasure precision* on the target concept and high *surgicality* on control concepts.

We argue that a central reason for the utility collapse of ambient-space erasure is geometric. The hidden space of a language model can be high-dimensional, but the image of actual sentence representations need not fill that ambient Euclidean space. Instead, it can concentrate near a much lower-dimensional support. In our primary nonlinear setting, this effective dimension is roughly an order of magnitude smaller than the ambient one. When gradient-based erasure is applied in the full hidden space while the data occupy only this lower-dimensional support, the risk of stepping off the natural representation manifold increases. Once that happens, even a direction that is locally predictive of the target concept can damage unrelated structure simply by pushing the representation into regions the model does not normally occupy.

This observation suggests a different design principle for nonlinear erasure: the edit should follow the local geometry of the representation manifold rather than the full ambient space. Intuitively, if the relevant representation support is modeled locally by a tangent space, then only the tangent component of an erasure update preserves the first-order geometry actually used by the model. Constraining edits to that tangent geometry should make erasure more surgical and, crucially, should allow larger erasure strengths λ before control concepts collapse.

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Our contributions are fourfold:

1. **A geometric formulation of nonlinear concept erasure.** We formalize the mismatch between ambient hidden dimension and effective representation dimension, and argue that off-manifold edits are a key failure mode of ambient gradient-based erasure.
2. **A tangent-constrained nonlinear erasure method.** In Section 3, we introduce a method that projects ambient erasure updates onto locally estimated tangent spaces so that interventions stay aligned with the representation geometry used by the model.
3. **An iterative erasure view motivated by recoverability.** Following Elazar and Goldberg's recoverability insight [Elazar and Goldberg, 2018], we do not treat success against a single scorer as evidence of complete erasure. Instead, the method is evaluated against freshly retrained nonlinear probes, and the erasure itself is iterated to remove residual recoverable signal.
4. **A supervised evaluation centered on precision and surgicality.** We evaluate on three text datasets and one large vision benchmark using retrained concept probes to measure erasure precision and control-concept performance to measure surgicality.

Empirically, the resulting tangent-constrained method achieves the strongest nonlinear erasure across sycophancy, gender, safety, and CelebA visual attributes while maintaining competitive control-concept preservation. On CelebA, it wins 36 of 40 attribute erasure tasks, reducing mean nonlinear leakage to 0.707 (vs. 0.839 for LEACE), enabled by its ability to safely scale to larger λ values where ambient methods cannot.

2 Problem Setup and Geometric Background

We first define the representation-editing problem, then introduce the geometric language used throughout the paper. Our notation follows the manifold viewpoint in Tu's introduction to smooth manifolds [Tu, 2011]: we use local coordinate maps, tangent spaces, and differentials only as local approximations, not as theorem-level guarantees about neural representations.

Language model and encoder. Let Σ^* denote the set of all strings over a vocabulary Σ. A language model p defines a distribution over Σ^*. We write $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ for the hidden representation at layer ℓ, where d is the hidden dimension. For a fixed layer ℓ (omitted when clear from context), the representation of string s is $\mathbf{x} = \mathrm{enc}_\ell(s) \in \mathbb{R}^d$.

Concepts and data. A binary concept is defined by a labeling function $\phi : \Sigma^* \to \mathcal{C}$, where $\mathcal{C} = \{0, 1\}$. Each sample i has representation $\mathbf{x}_i \in \mathbb{R}^d$ and concept label $y_i = \phi(s_i) \in \mathcal{C}$, with an optional control label z_i (e.g., profession for gender erasure). We write $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ for the dataset and $\mathcal{D}_c = \{i : y_i = c\}$ for the class-c subset, with $N_c = |\mathcal{D}_c|$.

The class-conditional representation is the random variable $X_c \sim \mathrm{enc}_\ell(s) \mid \phi(s) = c$, i.e., the distribution of hidden states for strings labeled c.

Concept scorer. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a two-layer MLP trained with binary cross-entropy on $\mathcal{D}$ to predict y from $\mathbf{x}$. Its per-sample gradient with respect to the input is $\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i) \in \mathbb{R}^d$, which points in the direction of steepest increase in concept probability at $\mathbf{x}_i$.

Representation manifold. Each layer representation $\mathbf{x} = \mathrm{enc}_\ell(s)$ is formally a point in the ambient hidden space $\mathbb{R}^d$, where d is the full hidden-state dimension. Following Tu [Tu, 2011], we model the representations that actually occur for real inputs as lying near a smooth embedded submanifold $\mathcal{M} \subset \mathbb{R}^d$. The manifold $\mathcal{M}$ is our mathematical model of the representation manifold. Its intrinsic dimension is denoted by m, with $m < d$ and typically much smaller than d in our settings, meaning that the local degrees of freedom used by the data are far fewer than the directions available in the ambient hidden space.

Now fix a point $\mathbf{x} \in \mathcal{M}$. Let $U \subset \mathcal{M}$ be a small local neighborhood around $\mathbf{x}$, and let

$$\varphi : U \to \Omega \subset \mathbb{R}^m \tag{1}$$

be a local coordinate map. Here Ω is an open set in the coordinate space $\mathbb{R}^m$. Intuitively, φ assigns m local coordinates to nearby points on the manifold. We choose the coordinates so that $\varphi(\mathbf{x}) = 0$, meaning that the point $\mathbf{x}$ is represented by the origin in local coordinates. The inverse map $\varphi^{-1} : \Omega \to U \subset \mathcal{M} \subset \mathbb{R}^d$ is the corresponding local parametrization back into the ambient hidden space.

The tangent space at $\mathbf{x}$ is the best local linear approximation to the manifold near $\mathbf{x}$. Formally, it is the m-dimensional linear space

$$T_{\mathbf{x}}\mathcal{M} = d(\varphi^{-1})_0(\mathbb{R}^m) \subset \mathbb{R}^d, \tag{2}$$

where $d(\varphi^{-1})_0$ is the differential of the local parametrization φ^{-1} at the origin. In ordinary ML language, this differential can be read as the Jacobian or best local linear approximation of φ^{-1} at 0. It maps an infinitesimal coordinate displacement in $\mathbb{R}^m$ to the corresponding first-order displacement in the ambient space $\mathbb{R}^d$. Because $\mathcal{M}$ is embedded in $\mathbb{R}^d$, we can treat $T_{\mathbf{x}}\mathcal{M}$ as an m-dimensional linear subspace of the ambient hidden space.

Ambient versus tangent directions. For any ambient vector $\mathbf{g} \in \mathbb{R}^d$ attached to the point $\mathbf{x}$, the local geometric picture suggests a decomposition

$$\mathbf{g} = \mathbf{g}^\top + \mathbf{g}^\perp, \qquad \mathbf{g}^\top \in T_{\mathbf{x}}\mathcal{M}, \quad \mathbf{g}^\perp \perp T_{\mathbf{x}}\mathcal{M}. \tag{3}$$

Here $\mathbf{g}^\top$ is the tangent component, namely the part of $\mathbf{g}$ that moves along the manifold, and $\mathbf{g}^\perp$ is the normal component, namely the part that points away from the manifold. In this local approximation, the tangent component changes the representation along directions already available in natural data, while the normal component is the first-order source of off-manifold displacement. This is the geometric distinction used later by TanGCE: ambient gradient updates may contain both parts, but tangent-projected updates keep only the component that remains aligned with the local geometry of plausible representations.

Recoverability after erasure. Throughout the paper, concept removal is judged only after training a *fresh* probe on the edited representations. This follows the lesson of Elazar and Goldberg [2018]: erasure succeeds only if a newly trained classifier still fails to recover the concept from $\tilde{\mathbf{x}}$.

Conventions. Index i ranges over samples. $\|\cdot\|$ denotes the Euclidean norm unless stated otherwise.

2.1 Baselines and Ambient Gradient Editing

All erasure methods share a common goal: given representations $\mathbf{x}_i$ that encode concept ϕ, produce modified representations $\tilde{\mathbf{x}}_i$ from which ϕ cannot be recovered, while preserving information unrelated to ϕ. We review the linear baselines and the ambient nonlinear baselines before introducing the tangent-constrained method in Section 3.

Linear projection. Given a concept direction $\mathbf{w} \in \mathbb{R}^d$ (typically the difference of class-conditional means, $\mathbf{w} = \mathbb{E}[X_1] - \mathbb{E}[X_0]$), linear projection removes the component along $\mathbf{w}$:

$$\tilde{\mathbf{x}}_i = (\mathbf{I} - \hat{\mathbf{w}}\hat{\mathbf{w}}^\top)\mathbf{x}_i, \qquad \hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|. \tag{4}$$

This removes a single linear direction but leaves all other concept-correlated structure intact.

INLP (Iterative Nullspace Projection). INLP [Ravfogel et al., 2020] iteratively trains linear classifiers $\mathbf{w}_1, \ldots, \mathbf{w}_T$ and projects onto the intersection of their nullspaces:

$$\mathbf{P}_{\text{INLP}} = \prod_{t=1}^{T}(\mathbf{I} - \hat{\mathbf{w}}_t\hat{\mathbf{w}}_t^\top), \qquad \tilde{\mathbf{x}}_i = \mathbf{P}_{\text{INLP}}\,\mathbf{x}_i. \tag{5}$$

Each $\mathbf{w}_t$ is trained on the residual after projecting out $\mathbf{w}_1, \ldots, \mathbf{w}_{t-1}$. INLP extends linear projection to multiple directions but remains limited to linearly decodable concept information.

LEACE (Least-squares Concept Erasure). LEACE [Belrose et al., 2023] finds an oblique projection $\mathbf{P} \in \mathbb{R}^{d \times d}$ of rank $|\mathcal{C}| - 1$ and a translation $\mathbf{b}$ that minimally modify representations while ensuring no log-linear classifier can recover the concept:

$$\tilde{\mathbf{x}}_i = \mathbf{P}\mathbf{x}_i + \mathbf{b}. \tag{6}$$

LEACE guarantees $\mathbb{E}[\tilde{X}_c] = \mathbb{E}[X]$ and $\mathrm{Var}[\tilde{X}_c] = \mathrm{Var}[\tilde{X}_{c'}]$ for all $c, c' \in \mathcal{C}$, erasing both mean and covariance differences between class-conditional distributions. This provides the strongest linear erasure guarantee among the baselines.

2.2 Gradient-Based Concept Erasure

The linear baselines above remove linearly decodable concept information but leave nonlinear structure intact. Several methods address this by computing per-sample erasure directions from the gradient of a nonlinear concept scorer.

IGBP (Iterative Gradient-Based Projection). IGBP [Iskander et al., 2023] trains a two-class MLP probe $g_\phi : \mathbb{R}^d \to \mathbb{R}^2$ with softmax output, then retracts each representation to the decision boundary via a Newton step. Let $p_i = \max_c \mathrm{softmax}(g_\phi(\mathbf{x}_i))_c$ be the maximum class probability. IGBP uses the *projection loss* $L_i = \frac{1}{2}\,\mathrm{logit}(p_i)^2$ and computes:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \frac{\nabla_{\mathbf{x}} L_i}{\|\boldsymbol{\theta}_i\|^2}, \qquad \boldsymbol{\theta}_i = \frac{\nabla_{\mathbf{x}} L_i}{\mathrm{logit}(p_i)}, \tag{7}$$

where $\mathrm{logit}(p) = \log \frac{p}{1-p}$. This is a Newton step that projects $\mathbf{x}_i$ onto the level set $\mathrm{logit}(p_i) = 0$ (i.e., $p_i = 0.5$, the decision boundary). Within each round, IGBP is therefore a one-shot boundary correction for the current scorer. The procedure then iterates: after projection, a fresh probe is trained on the projected representations and the process repeats until leakage falls to majority accuracy. Unlike Ambient GCE (below), IGBP has no explicit λ parameter—the step size is determined by each sample's distance to the decision boundary.

Ambient GCE. Ambient GCE is the tangent-free ablation used later to isolate the manifold constraint: instead of moving directly to the current decision boundary, it takes a tunable λ-scaled step along the scorer gradient. Given a trained scorer f_θ, the ambient erasure direction at sample i is the unit gradient:

$$\mathbf{u}_i = \frac{\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)}{\|\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)\|}. \tag{8}$$

The erased representation is obtained by projecting out the concept component:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\,(\mathbf{x}_i^\top \mathbf{u}_i)\,\mathbf{u}_i, \tag{9}$$

where $\lambda > 0$ controls the erasure strength. At $\lambda = 1$, this removes the full projection onto $\mathbf{u}_i$; at $\lambda < 1$, it partially attenuates.

Ambient GCE edits in the full ambient space and therefore does not distinguish tangent from normal directions. Section 3 introduces the tangent-constrained version of this update.

3 Method: TanGCE

We now introduce *Tangent Gradient-based Concept Erasure* (TanGCE), the method presented in this paper. The method is built from the geometric background above: if the representation support is locally approximated by an embedded manifold $\mathcal{M} \subset \mathbb{R}^d$, then the ambient erasure gradient should be edited so that the update moves along the tangent space $T_{\mathbf{x}}\mathcal{M}$ rather than along directions normal to the manifold.

3.1 Local tangent estimation

For each query point $\mathbf{x}_i$, we estimate the tangent space $T_{\mathbf{x}_i}\mathcal{M}$ via local PCA on its k-nearest neighbors. Let $\mathcal{N}_k(\mathbf{x}_i) = \{\mathbf{x}_{j_1}, \ldots, \mathbf{x}_{j_k}\}$ denote the k nearest neighbors of $\mathbf{x}_i$ in the training set. We form

secant vectors $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ and compute the local covariance:

$$\mathbf{C}_i = \frac{1}{k} \sum_{m=1}^{k} \mathbf{s}_{j_m} \mathbf{s}_{j_m}^\top \in \mathbb{R}^{d \times d}. \tag{10}$$

The top r eigenvectors of $\mathbf{C}_i$ form the tangent basis

$$\mathbf{B}_i = [\mathbf{b}_i^{(1)}, \ldots, \mathbf{b}_i^{(r)}] \in \mathbb{R}^{d \times r}, \tag{11}$$

where the intrinsic dimension r is selected automatically by a retained-variance rule. The resulting orthogonal projector onto the estimated tangent space is

$$\mathbf{P}_{T_i} = \mathbf{B}_i \mathbf{B}_i^\top. \tag{12}$$

This is the empirical analogue of the differential-geometric tangent space from Section 2: the neighborhood PCA basis provides a local linear model of the representation manifold around $\mathbf{x}_i$.

3.2 Tangent-constrained update

Given the ambient erasure direction $\mathbf{u}_i$ from Eq. 8, TanGCE defines the tangent-constrained direction by projecting onto the local tangent space:

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i} \mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i). \tag{13}$$

The projected direction is *not* re-normalized to unit length. Its norm $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as a confidence weight: when the ambient concept gradient is mostly normal to the manifold, $\|\mathbf{u}_i^{\tan}\| \ll 1$, and the edit is naturally attenuated.

TanGCE then applies the update

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda \left(\mathbf{x}_i^\top \mathbf{u}_i^{\tan}\right) \mathbf{u}_i^{\tan}. \tag{14}$$

The effective step size is $\lambda \|\mathbf{u}_i^{\tan}\|^2$, which is maximized when the gradient lies entirely in the tangent space and vanishes when it is entirely off-manifold.

This construction is the core difference between TanGCE and its ambient counterpart. Ambient GCE applies the update in the full hidden space, whereas TanGCE restricts the update to tangent directions supported by the local geometry of the representations.

3.3 Why tangent projection permits larger λ

Ambient GCE edits $\mathbf{x}_i$ along $\mathbf{u}_i \in \mathbb{R}^d$, which can contain a large normal component relative to $T_{\mathbf{x}_i}\mathcal{M}$. If $\mathbf{u}_i^\perp$ is large, increasing λ pushes the representation away from the local manifold approximation and can distort unrelated structure or preserve recoverable signal. TanGCE removes this normal component at first order by replacing $\mathbf{u}_i$ with $\mathbf{u}_i^{\tan}$. This geometric mechanism can permit operation at larger erasure strengths than ambient updates.

3.4 Iterative TanGCE and relation to baselines

We implement TanGCE using an iterative protocol with probe recycling. Each round begins by training a fresh concept scorer on the current representations, computing per-sample gradients, and applying the tangent-constrained update in Eq. 14. This turns the method into an iterative nonlinear eraser: every new scorer is trained specifically to recover whatever target information survived the previous round. The tangent estimator is also refit in every round to track the evolving local geometry.

The rationale follows Elazar and Goldberg [Elazar and Goldberg, 2018]: success against a single trained classifier is not evidence that the concept has been exhausted. We therefore evaluate every erasure method against a retrained probe after each full erasure procedure, and the erasure itself iterates to remove residual recoverable signal.

This setup also clarifies the relation between TanGCE and the baselines. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline, performing a one-shot correction to the current decision

boundary in each round. Ambient GCE is the direct ablation of TanGCE: it keeps the same fresh-scorer loop but applies the update in the full ambient space. TanGCE is the full method, adding the tangent-space restriction.

Two diagnostics verify tangent quality throughout the iterative process. *Tangent Gradient Capture* (TGC) measures the fraction of the leakage gradient lying within the estimated tangent space. *Tangent Neighbor Residual* (TNR) measures the fraction of local secant variation not explained by the tangent basis. High TGC and low TNR indicate that the local linear approximation is sufficiently faithful for the tangent-constrained update to be meaningful.

4 Evaluation

We evaluate each erasure method by training all probes from scratch on the erased representations, so that success cannot be attributed to leakage from the erasure scorer itself.

4.1 Metrics

Erasure precision. Let h^{NL} be a nonlinear probe and h^{L} a linear probe trained on $\tilde{\mathbf{x}}_i$ to predict the target concept y_i. We report their test accuracies on the erased representations:

$$\mathrm{Prec}^{\mathrm{NL}}_{\mathrm{erase}} = \mathrm{Acc}(h^{\mathrm{NL}}(\tilde{\mathbf{X}}), y), \qquad \mathrm{Prec}^{\mathrm{L}}_{\mathrm{erase}} = \mathrm{Acc}(h^{\mathrm{L}}(\tilde{\mathbf{X}}), y). \tag{15}$$

Lower values indicate stronger erasure. Because the target concepts in this paper are not assumed to be linearly encoded, the retrained nonlinear probe is the primary precision metric and the linear probe is a secondary diagnostic.

Surgicality on control concepts. Let q be a predictor for an annotated control label c (e.g., profession for gender erasure, helpfulness for safety, or selected low-correlation attributes in CelebA). We report the change in control performance relative to the clean representations:

$$\Delta_{\mathrm{ctrl}} = \mathrm{Acc}(q(\tilde{\mathbf{X}}), c) - \mathrm{Acc}(q(\mathbf{X}), c). \tag{16}$$

Smaller $|\Delta_{\mathrm{ctrl}}|$ means the erasure is more surgical. In CelebA, where each target attribute is paired with multiple controls, we report the mean Δ_{ctrl} across the selected controls.

Together, these metrics ask whether an erasure method removes the target concept while preserving unrelated control concepts.

4.2 Evaluation Protocol

Fresh-probe evaluation. The precision metric is intentionally strict: after every erasure method finishes, we train a *new* probe on the erased representations. This follows the cautionary lesson of Elazar and Goldberg [Elazar and Goldberg, 2018]: if a freshly trained probe can still recover the concept, then the concept was not actually removed in the sense relevant for downstream use.

Shared validation protocol. Most methods share a common erasure strength parameter λ (Eqs. 9 and 14). Because TanGCE's tangent projection attenuates the effective step size by $\|\mathbf{u}_i^{\tan}\|^2 \leq 1$, a fixed λ across methods would systematically under-erase for the tangent-constrained update. Rather than manually choosing different values, we adopt validation-based hyperparameter selection: all λ-based methods search the same grid $\lambda \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ and select the λ^* that minimizes validation nonlinear probe accuracy. IGBP has no explicit λ parameter—its step size is determined by each sample's distance to the decision boundary—so it runs until convergence.

Domain-specific controls. For sycophancy we use answer matching as the control concept; for Bias in Bios we use profession; for safety we use helpfulness; and for CelebA we use selected low-correlation facial attributes. This lets the evaluation distinguish targeted concept removal from broad representational damage.

Table 1: Comparison of concept erasure methods by implementation status and role in this work. We distinguish between methods implemented in this repository and those positioned as literature baselines.

Method	Family	Implemented	Replication	Canonical Dataset	Role in Paper
Linear Projection	Linear	Yes	Yes	Bias in Bios	Global baseline
INLP	Linear	Yes	Yes	Bias in Bios	Linear baseline
LEACE	Linear	Yes	Yes	Bias in Bios / Syc.	Matched linear baseline
IGBP	Nonlinear	Yes	Yes	Bias in Bios	Prior nonlinear baseline
`gce_ambient`	Nonlinear	Yes	Yes	Sycophancy	Nonlinear baseline
`gce_tangent`	Nonlinear	Yes	Yes	Sycophancy	Main contribution

5 Experiments

This section describes the empirical setup used to compare TanGCE to linear, ambient, and prior nonlinear baselines. The evaluation metrics and the shared validation protocol are defined separately in Section 4.

5.1 Experimental Domains

We evaluate our methods across four blocks:

1. **Sycophancy (Primary Nonlinear Case):** Our central focus is on removing agreement bias in model responses. This represents a critical challenge for model alignment where concept encoding is expected to be locally curved.
2. **Safety (Secondary Case Study):** We evaluate the surgicality of erasure on content safety using the PKU-SafeRLHF dataset, testing the preservation of helpfulness alongside safety leakage.
3. **Bias in Bios (Diagnostic Baseline):** We use gender and profession classification as a replication anchor. This setting provides a linear comparison point to contrast against the nonlinear gains of tangent-constrained erasure.
4. **CelebA Visual Attributes (Cross-Modal Generalization):** We apply the same erasure methods to CLIP ViT-B/32 `pooler_output` embeddings (768-dim) of CelebA face images. CelebA provides 40 binary facial attributes, enabling a rich surgicality analysis: when erasing one attribute (e.g., *Male*), the remaining 39 serve as potential controls. We select the 5 least-correlated attributes as controls per concept to ensure that control accuracy changes reflect genuine collateral damage rather than attribute correlation. This domain demonstrates that TanGCE generalizes beyond language model representations to vision encoders.

5.2 Models and Implementation

Language model. The sycophancy, safety, and gender experiments use Qwen2.5-0.5B (24 decoder layers, hidden dimension 896) at layer 8.

Vision encoder. The CelebA experiments use CLIP ViT-B/32 `pooler_output` embeddings (768-dim), extracted once and cached.

Methods. The core quantitative comparison evaluates TanGCE against six baselines and ablations: two random baselines (global and per-sample direction), linear projection, INLP [Ravfogel et al., 2020], LEACE [Belrose et al., 2023], and Ambient GCE. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline and serves as the main methodological reference point for the GCE family: it iteratively trains two-class MLP probes and retracts representations to the decision boundary via Newton steps (Eq. 7). Ambient GCE is our direct ablation of TanGCE: it uses the same scorer-training loop but applies the update in the full ambient space. TanGCE adds the key ingredient of the paper by estimating local tangent spaces with PCA-based neighborhoods and restricting the erasure direction to a local on-manifold approximation rather than the full ambient space.

Table 2: Full erasure method comparison at layer 8 across three concept domains. Each λ-based method's λ^* is selected from the same grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy, ensuring a fair comparison where every method gets its best shot at erasure; non-λ baselines use their standard procedure. L/NL Leak: retrained linear logistic-regression / nonlinear MLP probe accuracy on the target concept (lower = stronger erasure precision). ΔCtrl: downstream control-task accuracy change relative to clean representations (closer to zero = more surgical). Model: Qwen2.5-0.5B. Control tasks: answer matching (sycophancy), profession classification (gender), helpfulness (safety).

	Sycophancy			Gender (Bias in Bios)			Safety		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.685	.748	.000	.719	.758	.000	.798	.805	+.003
Random Per-Samp.	.621	.745	.000	.723	.743	+.003	.784	.799	$-.010$
Linear Proj.	.683	.740	.000	.722	.747	+.001	.807	.800	$-.003$
INLP	.683	.734	.000	.722	.737	+.001	.807	.795	$-.004$
LEACE	.681	.681	.000	.719	.739	$-.002$	.801	.785	$-.015$
Ambient GCE	.714	.703	.000	.663	.701	.000	.780	.778	$-.040$
TanGCE	**.628**	**.602**	$-.015$	**.603**	**.618**	$-.008$	**.654**	**.663**	+.039

5.3 Geometric Foundation (Appendix Support)

While the main text focuses on the performance of `gce_tangent`, we also provide geometric diagnostics to ground the mechanism story. These include broad geometry sweeps across multiple datasets and layers, subspace projection sensitivity (Q1), perturbation-based subspace sensitivity tests (Q2), and representation connectivity analyses (Q3). These appendix-scale materials support the low-dimensional and nonlinear-geometry motivation for tangent projection, but they should not be read as a standalone causal proof that off-manifold displacement fully explains the observed utility tradeoffs.

6 Results

We evaluate seven erasure methods across three concept domains—sycophancy, gender (Bias in Bios), and safety—at layer 8 of Qwen2.5-0.5B. Each method's erasure strength λ^* is selected from the same shared validation protocol described in Section 4. Our main metrics are erasure precision, measured by retrained probe accuracy on the target concept, and surgicality, measured by change in control-task performance.

6.1 TanGCE Achieves the Best Erasure Precision Across All Domains

Table 2 shows that TanGCE achieves the lowest nonlinear leakage across all three domains: 0.602 on sycophancy (vs. 0.681 for LEACE, 0.703 for Ambient GCE), 0.618 on gender (vs. 0.701 for Ambient GCE), and 0.663 on safety (vs. 0.778 for Ambient GCE). Linear methods (Linear Projection, INLP) barely reduce leakage below random baselines, confirming that concept information is predominantly encoded nonlinearly.

6.2 TanGCE Remains Surgical on Control Concepts

Table 2 reports ΔCtrl, the change in downstream control task accuracy relative to clean representations. For sycophancy (answer matching), TanGCE incurs a small -1.5pp drop; all other methods preserve perfect accuracy. For gender (profession classification), TanGCE drops only -0.8pp (0.531 vs. 0.539 clean). On safety, TanGCE *improves* helpfulness ($+3.9$pp), suggesting that removing safety-concept directions frees capacity for the helpfulness task.

6.3 The Lambda Sweep: On-Manifold Updates Support Larger Interventions

Table 3 shows the central operating-point difference between ambient and tangent erasure. As λ increases, Ambient GCE becomes unstable: its nonlinear leakage worsens sharply at high step sizes, rising to 0.734 on sycophancy, 0.702 on gender, and 0.770 on safety at λ=8. TanGCE behaves in the

Table 3: λ sweep comparing Ambient GCE and TanGCE at matched λ values ($\lambda \in \{0.5, 4.0, 8.0\}$) at layer 8. NL: retrained nonlinear probe accuracy on the target concept (lower = stronger erasure precision). Ambient GCE deteriorates at high λ, while TanGCE continues improving, supporting the claim that on-manifold updates can safely employ larger erasure strengths.

Concept	Method	λ	NL↓
Syco.	Ambient	0.5	.678
	TanGCE	0.5	.662
	Ambient	4.0	.694
	TanGCE	4.0	.596
	Ambient	8.0	.734
	TanGCE	8.0	**.588**
Gender	Ambient	0.5	.666
	TanGCE	0.5	.704
	Ambient	4.0	.700
	TanGCE	4.0	.580
	Ambient	8.0	.702
	TanGCE	8.0	**.470**
Safety	Ambient	0.5	.664
	TanGCE	0.5	.748
	Ambient	4.0	.776
	TanGCE	4.0	.680
	Ambient	8.0	.770
	TanGCE	8.0	**.646**

Table 4: CLIP ViT-B/32 concept erasure on CelebA (validation HPO for λ). Each method's λ^* is selected from the same grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy, ensuring a fair comparison. L/NL Leak: retrained linear / nonlinear probe accuracy on test (lower = better). ΔCtrl: mean control-task accuracy change across 5 least-correlated attributes (closer to zero = more surgical). Representation: CLIP ViT-B/32 `pooler_output` (768-dim).

		Male			Smiling			Blond_Hair		
Method	λ^*	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	0.5	.997	.990	-0.3	.903	.903	-0.3	.937	.923	-0.3
Random Per-Samp.	0.5	.993	.990	-0.1	.867	.887	-1.3	.933	.920	$+0.1$
Linear Proj.	1.0	.987	.987	-0.7	.903	.913	$+0.1$	.923	.940	-1.4
INLP	1.0	.587	.987	-0.2	.580	.740	$+0.6$	.713	.917	$+0.7$
LEACE	0.5	.997	.997	-0.3	.503	.820	$+0.4$	.903	.917	-0.1
Ambient GCE	0.5	.997	.997	-0.9	.787	.877	$+0.1$	.910	.923	-1.1
TanGCE	**8.0**	**.800**	**.847**	-1.1	**.487**	**.443**	-0.1	**.693**	**.720**	-1.6

opposite way, improving as λ increases and reaching its best precision at the largest tested step size in all three domains.

This pattern is consistent with the manifold hypothesis motivating the method. If the ambient gradient has a substantial off-manifold component, then increasing λ does not simply erase more target information; it also pushes the representation into regions where unrelated structure is distorted. Projecting onto the local tangent approximation suppresses those off-manifold components, which is why TanGCE can safely use larger erasure strengths.

6.4 Cross-Modal Generalization: CelebA Visual Attributes

To test whether TanGCE generalizes beyond language model representations, we apply all erasure methods to CLIP ViT-B/32 embeddings of CelebA face images. CelebA provides 40 binary facial attributes, giving a rich surgicality profile: when erasing one attribute, the remaining attributes serve as controls.

Table 5: CelebA all-40-attribute summary (CLIP ViT-B/32, validation HPO for λ). Each cell averages over all 40 binary CelebA attributes. λ^* is selected per method per concept from the grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL probe accuracy. NL Leak: mean nonlinear test accuracy across 40 concepts (lower = better). Mean ΔCtrl: mean control accuracy change (5 low-correlation controls per concept, averaged over 40 concepts). Worst ΔCtrl: worst single control drop averaged over 40 concepts.

Method	NL Leak↓	Mean ΔCtrl	Worst ΔCtrl	Med. λ^*
Random Global	0.883	-0.4%	-1.6%	2.0
Random Per-Samp.	0.882	-0.3%	-1.2%	0.5
Linear Proj.	0.869	-0.8%	-2.7%	4.0
INLP	0.859	-3.5%	-6.9%	1.0
LEACE	0.839	-0.2%	-1.2%	2.0
Ambient GCE	0.875	-0.6%	-2.8%	0.5
TanGCE	**0.707**	-0.7%	-2.1%	**8.0**

Table 6: Geometry contrast between sycophancy and gender across completed layers 4, 8, and 12. Lower intrinsic dimension and larger local-vs-global angle indicate stronger nonlinearity: a concept whose erasure direction varies significantly across the input space, making a single fixed direction insufficient to capture the full concept. Sycophancy shows substantially higher nonlinearity (dim 34–53, angle 76.9–82.6°) compared to gender (dim 365–461, angle 51.8–55.8°), supporting sycophancy as the primary case for manifold-aware erasure methods. This table demonstrates the geometric contrast but does not establish causality: it does not prove that the observed nonlinearity will be converted into lower displacement or higher utility preservation by any particular erasure method.

Concept	Local intrinsic dimension	Mean local/global angle
Sycophancy	34–53	76.9–82.6°
Gender	365–461	51.8–55.8°

Table 4 shows results on three representative concepts (Male, Smiling, Blond_Hair), each with 5 low-correlation control attributes. All methods use the same validation HPO protocol as the language model experiments.

Table 5 summarizes results averaged over all 40 CelebA attributes. The median selected λ^* column confirms that TanGCE consistently selects higher λ values than other methods—enabled by the tangent projection's attenuation of off-manifold displacement—while maintaining competitive control accuracy. Per-concept breakdowns for all 40 attributes (nonlinear leakage and mean ΔCtrl per method) are reported in Appendix A.4, Tables 10 and 11.

6.5 Geometric Evidence and Off-Manifold Displacement

Table 6 demonstrates that sycophancy is much more nonlinear than gender: its intrinsic dimension (34–53 across layers 4, 8, 12) is far lower than gender's (365–461), and its local-vs-global angle is much larger (76.9–82.6° vs. 51.8–55.8°). This geometry contrast is consistent with sycophancy being the primary case for manifold-aware erasure methods, while the strong TanGCE results on gender demonstrate that the method generalizes even to approximately linear concepts.

Appendix-Supporting Evidence. Detailed geometric analyses supporting these claims are provided in the Appendix, including:

- **Subspace Sufficiency (Q1):** Evidence that task-specific computation is concentrated in a low-dimensional linear subspace.
- **Perturbation Sensitivity (Q2):** Comparison of perturbations in high-variance and orthogonal subspaces.
- **Representation Connectivity (Q3):** Analysis of the connected, approximately convex topology of the representation manifold.

- **Multi-Layer Accumulation:** Results on how projection error accumulates across the network.

7 Discussion

The completed paper suite already narrows the central claim. In every finished comparison, the tangent-based method removes more measured concept signal than its ambient counterpart. That pattern appears in the sycophancy sweeps, in the completed layer-8 safety sweep, and in the Bias in Bios diagnostic. What the current paper snapshot does not yet show is the cleaner second half of the original thesis in which tangent projection also achieves lower displacement or uniformly better utility retention. In the completed runs, tangent reaches lower leakage by taking larger steps, and its utility metrics are either slightly worse or mixed.

The geometry diagnostics still matter because they explain why sycophancy is the harder case. Sycophancy shows local-vs-global angles near 80°, while gender remains near 53°, so a single global erasure direction is a much worse summary for the primary nonlinear concept. That is consistent with tangent-based methods improving removal in both concepts while leaving sycophancy as the clearly more nonlinear case, but the finished results currently frame tangent as a stronger remover rather than a dominantly better Pareto point.

The appendix snapshot tables slightly sharpen the near-term agenda. In the broader repository reports (Tables 7 and 8), the same removal advantage continues at layers 16 and 23 for both sycophancy and safety, although the utility metrics differ by domain and report. We do not elevate those rows to headline claims because they were not part of the paper-locked evidence set, but they strengthen the case for spending the next round of effort on matched-leakage comparisons and deeper-layer replications rather than on searching for a different primary mechanism.

Two implications follow. First, the paper should presently avoid claiming that off-manifold control is already proven to improve the full leakage-utility frontier; the matched-leakage analysis and broader replications are still needed for that claim. Second, the current evidence suggests that the next important question is not whether tangent projection changes the erasure direction at all, but under which layers and operating points that removal advantage can be converted into a cleaner utility tradeoff.

The matched layer-8 linear-family comparison clarifies one part of that tradeoff. LEACE is more surgical than tangent on this slice because it removes linearly decodable concept signal while taking a much smaller geometric step and preserving more downstream utility, especially on sycophancy where the answer-matching penalty is substantially smaller. That does not make LEACE a substitute for tangent on the main nonlinear claim: LEACE succeeds by solving the linear erasure problem cleanly, whereas tangent remains the stronger method when the target concept is evaluated with a fresh nonlinear probe. The current evidence therefore suggests a division of labor rather than a single winner: LEACE is the cleaner linear baseline, while tangent remains the more aggressive nonlinear remover.

8 Limitations

The current manuscript deliberately limits its main-text quantitative claims to the paper runs that were complete when this draft was assembled. That keeps the claims auditable, but it also means the strongest version of the utility-preservation story is not yet established in the locked evidence set. The deeper iterative sycophancy and safety reports are directionally consistent with the main narrative and are included in Appendix A, yet the paper's central claims still rely on the smaller completed suite.

A second limitation is measurement scope. All experiments use Qwen2.5-0.5B, and utility is measured through task-specific proxies such as answer matching, helpfulness ranking, profession classification, and dataset-source retention. These metrics are appropriate for surgicality, but they do not yet establish broad capability preservation across unrelated tasks.

Finally, the geometry diagnostics support low-dimensional structure, concept nonlinearity, and connectedness, but they do not by themselves prove that curved-manifold effects are the sole causal mechanism behind every leakage–utility tradeoff. The current paper therefore treats geometry as strong supporting evidence for tangent projection rather than as a complete causal proof.

9 Conclusion

We introduced TanGCE, the method presented in this paper: a manifold-aware concept erasure method that constrains erasure gradients to the local tangent space of the representation manifold. The central claim is not just that tangent projection removes more target information, but that it does so for the right geometric reason: when sentence representations occupy a much smaller natural manifold than the ambient hidden space, restricting updates to the tangent space lets the method use larger erasure strengths while keeping control-task changes small.

The comparison to IGBP and Ambient GCE clarifies the role of the manifold constraint. IGBP is the closest prior nonlinear baseline; Ambient GCE is the direct ablation with the same iterative loop but no tangent restriction; and TanGCE is the full method. The resulting λ sweep makes the practical benefit clear: ambient updates become unstable at large step sizes, whereas tangent-constrained updates continue to improve erasure precision.

Cross-modal experiments on 40 CelebA facial attributes in CLIP ViT-B/32 representations confirm that TanGCE generalizes beyond language models. TanGCE wins 36 of 40 attribute erasure tasks under fair HPO, reducing mean nonlinear leakage to 0.707 (vs. 0.839 for LEACE, the strongest baseline), with comparable control-task preservation. The remaining concepts are near-chance or favor another baseline.

More broadly, the iterative nonlinear protocol follows a conservative lesson from prior erasure work: if a freshly trained classifier can still recover the concept, then the concept is still present. By repeatedly retraining the scorer and constraining each erasure step to the tangent space, TanGCE turns that lesson into a practical nonlinear erasure method.

Several directions remain open. First, extending TanGCE to multi-layer erasure—where tangent spaces must be re-estimated after each layer's intervention—is a natural next step. Second, while our experiments use PCA-based tangent estimation, learned manifold representations (e.g., variational autoencoders) could provide more accurate tangent spaces. Finally, the connection between local geometric structure and erasure effectiveness suggests that manifold-aware methods may benefit other representation editing tasks beyond concept removal.

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Table 7: Repository snapshot from the iterative sycophancy report at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is answer-matching delta relative to clean representations. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This snapshot table is not a main-text claim anchor and should be read only as evidence of directional consistency with the layer-8 paper-locked row (Table 2). Deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines across this exploratory sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.063	0.213	-20.7pp	-0.5pp	2.88	0.62
16	0.153	0.400	+0.0pp	+0.0pp	3.79	0.27
23	0.138	0.361	-0.1pp	-0.2pp	44.50	3.11

Table 8: Repository snapshot from the iterative safety report at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is pairwise helpfulness delta. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This snapshot table is not a main-text claim anchor and should be read only as evidence of directional consistency with the layer-8 paper-locked row (Table 2). Deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines across this exploratory sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.0725	0.2583	-0.052	-0.023	2.70	0.18
16	0.1292	0.3517	-0.092	+0.003	4.37	0.60
23	0.1483	0.3404	-0.062	-0.152	38.32	5.30

A Repository Snapshot Tables

This appendix records broader repository snapshot tables that are directionally consistent with the main text but were not part of the locked paper-evidence set when the quantitative claims in Section 6 were written. We include them to document the current state of the project and to guide the next round of matched-leakage and deeper-layer analyses. The central paper claims continue to rely only on the completed paper suite summarized in the main text.

A.1 Iterative Sycophancy Snapshot

A.2 Iterative Safety Snapshot

A.3 Local CelebA Visual Table-2-Style Comparison

To ensure a modality-matched benchmark, this appendix now reports a vision-only CelebA run (CLIP ViT-B/32 embeddings) with the Table 2 method family and 3 least-correlated controls per target concept. Artifacts are stored under `outputs/local/celeba_visual_table2_v1/` (including per-method sample folders with 2 saved images each). The consolidated W&B results run is `g52h8kbr`.

A.4 CelebA Per-Concept Detail (All 40 Attributes)

Tables 10 and 11 give the full per-concept breakdown behind the aggregate CelebA result in Table 5. The main-text Table 4 reports three representative concepts (Male, Smiling, Blond_Hair); the appendix tables cover all 40 binary CelebA attributes from the same `celeba_clip_all40_fair_v2` run under the shared validation-HPO protocol. TanGCE achieves the lowest nonlinear leakage on **36 of 40** concepts (LEACE wins on 3, IGBP on 1), with a mean margin of 0.116 absolute NL accuracy over the best competing method per concept. Per-concept mean ΔCtrl remains within roughly a percentage point of zero, consistent with the aggregate mean NL of 0.707 and mean ΔCtrl of -0.7% reported in the main text.

Geometry Note. The broader geometry report continues to support three useful points: early and mid-layer representations are effectively low-dimensional, on-manifold directions carry much

Table 9: Local Table-2-style CelebA visual comparison using CLIP ViT-B/32 embeddings. For each target concept we intervene once per method and evaluate (i) concept leakage via re-trained linear/nonlinear probes (lower is better) and (ii) mean control-task accuracy change across 3 least-correlated control attributes (closer to zero is better). Targets and controls: Male $\rightarrow$ Blurry,Narrow_Eyes,Straight_Hair; Smiling $\rightarrow$ Straight_Hair,Bushy_Eyebrows,Gray_Hair; Blond_Hair $\rightarrow$ Straight_Hair,Narrow_Eyes,Blurry. Lambdas follow Table 2: all methods λ=0.5 except TanGCE λ=8.0.

	Male			Smiling			Blond_Hair		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.993	.994	-0.0pp	.891	.911	+0.0pp	.909	.933	-0.1pp
Random Per-Samp.	.993	.994	+0.0pp	.891	.907	+0.2pp	.907	.932	-0.1pp
Linear Proj.	.993	.992	+0.0pp	.900	.906	-0.3pp	.919	.931	-0.2pp
INLP	.994	.993	-0.0pp	.900	.903	+0.3pp	.918	.932	-0.0pp
LEACE	.994	.991	-0.0pp	.894	.894	+0.2pp	.909	.925	-0.1pp
Ambient GCE	.990	.995	-0.2pp	.858	.884	-0.4pp	.893	.920	+0.2pp
TanGCE	.887	.953	-0.8pp	.649	.649	-0.6pp	.699	.747	-2.1pp

of the model's sensitivity in these perturbation tests, and interpolation remains consistent with a connected representation space. At the same time, the Q2 perturbation study is best read as evidence of sensitivity concentration in a high-variance subspace rather than as a standalone proof of curved-manifold causality.

Table 10: Per-concept nonlinear probe leakage on CelebA (CLIP ViT-B/32), all 40 binary attributes. Each cell is retrained nonlinear probe test accuracy after erasure (lower = stronger erasure). λ^* is selected per method per concept from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL accuracy. Bold marks the lowest leakage per row. Abbrev.: RandG = Random Global, RandS = Random Per-Sample, LinProj = Linear Projection, AmbGCE = Ambient GCE, TanGCE = Tangent GCE.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	.92	.92	.91	.92	.87	.92	.91	**.77**
Arched_Eyebrows	.79	.79	.76	.76	.69	.77	.77	**.59**
Attractive	.73	.74	.69	.73	.63	**.53**	.73	.60
Bags_Under_Eyes	.76	.75	.74	.75	.70	.77	.75	**.69**
Bald	1.00	.99	.99	.99	.98	1.00	.99	**.97**
Bangs	.91	.91	.89	.89	.87	.92	.91	**.68**
Big_Lips	.80	.79	.74	.80	**.68**	.82	.81	.73
Big_Nose	.76	.76	.73	.74	.72	.73	.76	**.61**
Black_Hair	.85	.84	.82	.81	.79	.83	.82	**.46**
Blond_Hair	.92	.92	.94	.92	.92	.92	.92	**.72**
Blurry	.98	.98	.98	.96	.97	.98	.97	**.88**
Brown_Hair	.80	.80	.76	.75	.75	.80	.80	**.52**
Bushy_Eyebrows	.87	.88	.86	.85	.76	.87	.86	**.66**
Chubby	.94	.94	.95	.95	.92	.95	.94	**.81**
Double_Chin	.94	.94	.94	.93	.92	.94	.95	**.85**
Eyeglasses	.99	1.00	.96	.93	.96	1.00	.96	**.90**
Goatee	.96	.97	.97	.93	.96	.96	.95	**.85**
Gray_Hair	.97	.97	.97	.97	.97	.97	.98	**.89**
Heavy_Makeup	.89	.89	.89	.86	.85	.88	.88	**.70**
High_Cheekbones	.79	.79	.77	.75	.70	.57	.76	**.50**
Male	.99	.99	.99	.99	1.00	.98	1.00	**.85**
Mouth_Slightly_Open	.89	.89	.89	.71	.72	.66	.80	**.43**
Mustache	.94	.95	.95	.94	.92	.95	.93	**.77**
Narrow_Eyes	.88	.88	.80	.87	.80	.88	.88	**.78**
No_Beard	.94	.95	.94	.93	.92	.94	.95	**.77**
Oval_Face	.70	.71	.69	.68	**.67**	.69	.67	.72
Pale_Skin	.93	.93	.94	.95	.93	.92	.93	**.83**
Pointy_Nose	.75	.72	.71	.69	**.63**	.72	.75	.64
Receding_Hairline	.94	.94	.96	.96	.94	.94	.95	**.69**
Rosy_Cheeks	.93	.93	.93	.94	.91	.94	.93	**.79**
Sideburns	.95	.95	.94	.94	.95	.94	.94	**.82**
Smiling	.90	.89	.91	.74	.82	.59	.88	**.44**
Straight_Hair	.78	.78	.79	.77	.78	.78	.78	**.49**
Wavy_Hair	.78	.76	.77	.77	.78	.77	.78	**.52**
Wearing_Earrings	.90	.89	.86	.88	.85	.89	.88	**.61**
Wearing_Hat	1.00	1.00	.97	.97	.97	1.00	.97	**.94**
Wearing_Lipstick	.90	.90	.90	.89	.85	.88	.92	**.75**
Wearing_Necklace	.84	.85	.80	.80	.77	.85	.83	**.68**
Wearing_Necktie	.93	.93	.92	.92	.91	.92	.93	**.78**
Young	.86	.86	.85	.84	.83	.86	.87	**.60**
Mean	.88	.88	.87	.86	.84	.86	.87	**.71**

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Table 11: Per-concept mean control accuracy change ΔCtrl (%) on CelebA, all 40 binary attributes. For each concept, ΔCtrl is averaged over the 5 least-correlated control attributes (closer to zero = more surgical). λ^* selection matches Table 10.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	+0.0	+0.1	-0.7	+0.1	+0.0	+0.1	-2.3	-0.2
Arched_Eyebrows	-1.3	-2.3	-1.2	-0.8	-1.1	-2.5	+0.6	-0.6
Attractive	-0.3	-1.0	-1.7	-0.8	-0.3	-0.9	+0.0	-0.8
Bags_Under_Eyes	-3.7	-1.7	-0.3	-7.7	-0.3	-0.3	-0.3	-0.1
Bald	+0.1	+0.0	-0.1	+0.1	-0.1	+1.1	+2.3	+0.2
Bangs	-3.7	-1.2	-0.5	-7.9	-0.1	-0.7	-0.3	-0.8
Big_Lips	+0.1	-0.0	-0.3	+0.0	-0.1	-0.5	+0.1	-0.1
Big_Nose	-0.1	+0.3	-3.5	-9.7	+0.0	-0.3	-1.8	+0.1
Black_Hair	-0.1	-0.0	+0.1	+0.2	+0.3	-0.3	-0.3	-0.7
Blond_Hair	-0.3	+0.1	-1.4	+0.7	-0.1	-0.4	-1.1	-1.6
Blurry	-1.3	+0.0	-0.1	-5.5	+0.1	-0.7	+0.7	-0.1
Brown_Hair	+0.1	-0.9	-0.3	-6.0	-0.7	+0.4	+0.1	-0.2
Bushy_Eyebrows	+0.0	+0.1	-0.1	-1.5	+0.2	-0.9	-0.3	-0.5
Chubby	-0.1	+0.3	-0.2	-1.4	-0.2	-0.2	+0.4	-2.0
Double_Chin	+0.1	+0.1	-0.4	-12.7	+0.3	+0.7	-0.7	-2.5
Eyeglasses	-0.1	+0.1	-0.0	-0.1	-0.2	-0.4	-0.2	+0.2
Goatee	+0.1	-0.1	+0.0	+0.2	+0.1	+0.5	-0.3	-0.4
Gray_Hair	-0.2	-0.3	-2.1	+0.4	-0.1	-1.1	-1.5	-1.5
Heavy_Makeup	-0.3	-0.1	-1.1	-1.7	-1.3	-1.3	-1.1	-0.9
High_Cheekbones	+0.1	+0.2	+0.1	-0.3	-0.1	-0.4	-0.0	-0.7
Male	-0.3	-0.1	-0.7	-0.2	-0.3	-2.2	-0.9	-1.1
Mouth_Slightly_Open	-0.3	-0.6	+0.1	+0.4	+0.2	+0.3	+0.1	-0.2
Mustache	+0.0	-0.1	+0.1	-10.3	-0.5	+1.0	+0.5	-1.5
Narrow_Eyes	-0.1	-0.2	-0.3	+0.2	-0.1	+0.3	+0.0	+0.0
No_Beard	-1.2	-0.2	-4.0	-6.5	-1.6	-0.1	-0.7	-0.5
Oval_Face	-2.0	+0.1	-0.1	-14.4	-0.6	-0.3	+0.2	-0.3
Pale_Skin	-1.2	-0.1	-0.1	-6.8	-0.9	-0.1	+0.7	-1.0
Pointy_Nose	-0.1	-0.2	+0.3	-1.1	+0.2	-0.7	-0.1	-0.1
Receding_Hairline	-0.1	-0.1	-0.1	-5.9	+0.0	-0.3	+0.1	-1.1
Rosy_Cheeks	-0.5	+0.1	-0.6	-0.1	-0.2	-0.1	-0.1	-1.3
Sideburns	-0.4	+0.0	-0.1	+0.1	+0.1	-0.6	-2.5	-0.1
Smiling	-0.3	-1.3	+0.1	+0.6	+0.4	+0.1	+0.1	-0.1
Straight_Hair	-0.1	-0.2	-0.1	-0.9	-0.1	-0.8	+0.5	-0.7
Wavy_Hair	+0.1	-0.7	-0.9	+0.2	+0.1	-0.4	-0.1	-0.7
Wearing_Earrings	+0.8	-0.3	-1.6	-0.3	-0.1	-1.3	-0.5	-0.3
Wearing_Hat	-0.1	+0.0	-0.1	-0.6	+0.0	+1.0	+0.1	-0.2
Wearing_Lipstick	+0.2	-0.1	-6.1	-21.5	-1.9	-1.2	-1.4	-0.9
Wearing_Necklace	+0.1	-1.7	-1.7	-9.3	+0.3	-1.8	+0.2	-1.2
Wearing_Necktie	+0.0	+0.1	+0.0	+0.2	-0.2	-0.7	-13.4	-1.3
Young	-0.0	-0.7	-1.7	-10.7	+0.1	-0.9	-2.2	-0.8
Mean	-0.4	-0.3	-0.8	-3.5	-0.2	-0.4	-0.6	-0.7

Table 12: Geometry evidence map organizing the appendix diagnostics into four research blocks. Each block addresses a distinct aspect of manifold structure: Q1 examines whether task-relevant computation concentrates in a low-dimensional linear subspace, Q2 tests whether perturbations on the manifold are more consequential than off-manifold perturbations, Q3 validates whether the manifold is connected across domains, and multi-layer accumulation tracks error compounding through the network. These diagnostics support the mechanism discussion and motivate the need for manifold-aware erasure methods but do not by themselves establish causality.

Experiment Block	Research Question	Primary Evidence
Q1: Rank-retention	Subspace concentration	Top-1 agreement under intrinsic projection
Q2: Perturbation sensitivity	High-variance vs. orthogonal directions	KL and top-1 under on/off-subspace noise
Q3: Domain connectivity	Shared vs. domain manifolds	Cross-domain interpolation smoothness
Multi-layer accumulation	Error compounding across the network	Simultaneous intrinsic projection across layers